

KHAJA MUBASHIRUDDIN

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PROFESSIONAL SUMMARY

AI and Full Stack Engineer delivering production AI Systems in Healthcare and SaaS. Designed and shipped end-to-end machine learning pipelines spanning supervised learning, CNNs, and LLM-powered applications – from raw data ingestion and feature engineering through model training, evaluation, and interactive visualization. Comfortable operating across the full stack: Python ML backends, Flask/Node APIs, and accuracy models with measurable downstream impact – including a 15% latency reduction in production, a 40% improvement in clinician report usability, and a 94% gesture recognition accuracy on a custom deep learning system. Currently seeking AI/ML engineering or applied ML research roles where technical dept and cross-functional execution both matter

EXPERIENCE.

Front-End Engineer Intern – Lunacal.ai | Remote | July 2024 – Sept 2024

- Reduced page load time by 15% by auditing and refactoring theme performance in React/Next Js component library, validated via Lighthouse benchmarking.
- Shipped 8+ responsive calendar UI components with pixel-perfect fidelity to Figma specs, all components adopted into production within 2 weeks.
- Reduced average component review cycle from 3 days to 1 day by establishing a shared Figma-to-code handoff convention across the frontend team.
- Prototyped 2 AI-enhanced scheduling features during a 2-week sprint; both were escalated to the core product roadmap after intern day presentation.
- Collaborated cross-functionally with 3 designers and 2 backend engineers to align dynamic calendar interfaces with live API data constraints.

Software Engineering Intern – Inaara.ai | Remote | May 2024 – July 2024

- Built AI-powered clinical dashboards that improved report clarity and usability scores by 40%, measured via structured clinician feedback surveys.
- Reduced average data-to-visualization pipeline time by ~30% by redesigning backend data-presentation workflows in Flask and restructuring MongoDB query patterns.
- Decreased average clinician time-on-task for reading reports by an estimated 25% by introducing progressive data loading and collapsible insight panels.
- Conducted end-to-end UI/UX testing across 6 core user flows, identifying and resolving 12 friction-point issues before public launch.
- Delivered all 3 assigned feature modules ahead of schedule, enabling the engineering team to pull forward a planned Q3 feature into the Q2 release cycle.

PROJECTS

SleepSense – AI-Driven Student Wellness & GPA Prediction – (UB CSE-587 | Runner-Up, Best Project Award)

- Engineered an end-to-end ML pipeline on wearable data (Sleep, HR, SpO₂, Stress) from 100+ student records to predict GPA outcomes with interpretable outputs.
- Trained and benchmarked 4 supervised models (Linear Regression, SVM, Random Forest, KNN); Random Forest achieved the best F1-score of 0.86 after a 5-fold cross-validation.
- Applied targeted feature engineering and noise-handling techniques on real-world wearable data, improving model accuracy by ~18% over unprocessed baseline.
- Built interactive NextJS dashboards visualizing per-student psychological trends and model confidence scores, used for final presentation to faculty panel of 5 judges.
- Stack: Python, Scikit-learn, Pandas, Matplotlib, Next JS, MongoDB

GestureDrive – Real-Time In-Car Gesture Recognition System – (UB CSE-676 Deep Learning)

- Trained CNN models on HaGrid Dataset (~552K images, 18 gesture classes) for static and dynamic hand gesture detection, achieving 94% top-1 accuracy on held-out test set
- Optimized inference pipeline to run under 100ms latency constraint, enabling real-time gesture control simulation within a React/Next JS in-cat infotainment inference.
- Improved model generalization by 11% over initial baseline by applying data augmentation (random crops, flips, color jitter) and dropout regularization.
- Reduced false positive rate to under 4% across all gesture class by fine-tuning confidence thresholds per class using precision-recall curve analysis.
- Integrated OpenCV preprocessing layer to normalize lighting and hand position variance across 6 distinct recording environments in the dataset.
- Stack: Python, PyTorch, OpenCV, React, Next JS, Tailwind CSS

EDUCATION

Masters (M.S.) in Computer Science – AI/ML Track

University at Buffalo (SUNY) | Jan 2025 – May 2026

- Competed against ~120 students in class-wide project showcase; awarded Runner-Up, Best Project (CSE 487/587) for SleepSense ML pipeline out of 40+ submitted projects.
 - Maintaining strong academic standing while concurrently developing two project systems, one of which (GestureDrive) reached 94% production accuracy.
- Coursework: Machine Learning, Deep Learning, Data Intensive Computing, Algorithms, Computer Vision

Bachelor of Technology (B.Tech) in Computer Science

Jawaharlal Nehru Technological University, Hyderabad (JNTUH) | 2020 – 2024

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, C++, Java

AI/ML: Scikit-Learn, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib – supervised learning, CNNs, model evaluation, feature engineering.

Web & APIs: React, Next JS, Express JS, Flask, REST APIs, MongoDB, Sanity, Stripe API, Google Fit API

Frontend / UI: Tailwind CSS, Framer Motion, Figma, responsive design, component architecture

Tools: Git, GitHub, Docker, Power BI, Jupyter Notebook, VS Code.